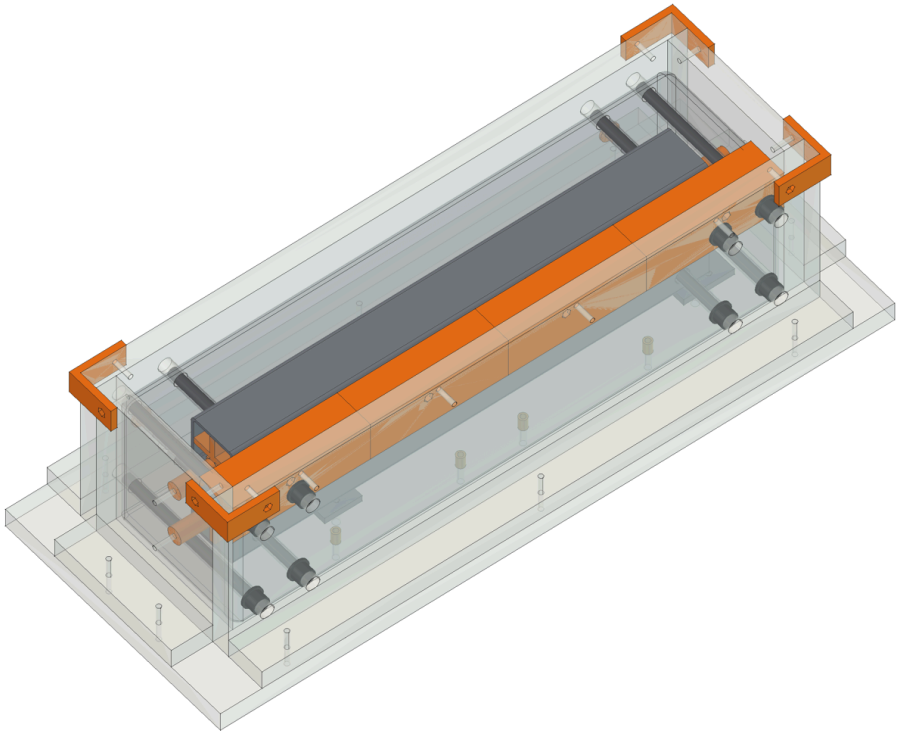
Beam mold

DIY CNC epoxy-granite mill · October 2026

How to make, put together and take apart the mold for the beam. The first section (how the molds work) and the last three (pouring, taking it apart, fixes) are the same in every mold guide; molds-complete.pdf has all the molds and the sheet layout for the boards.

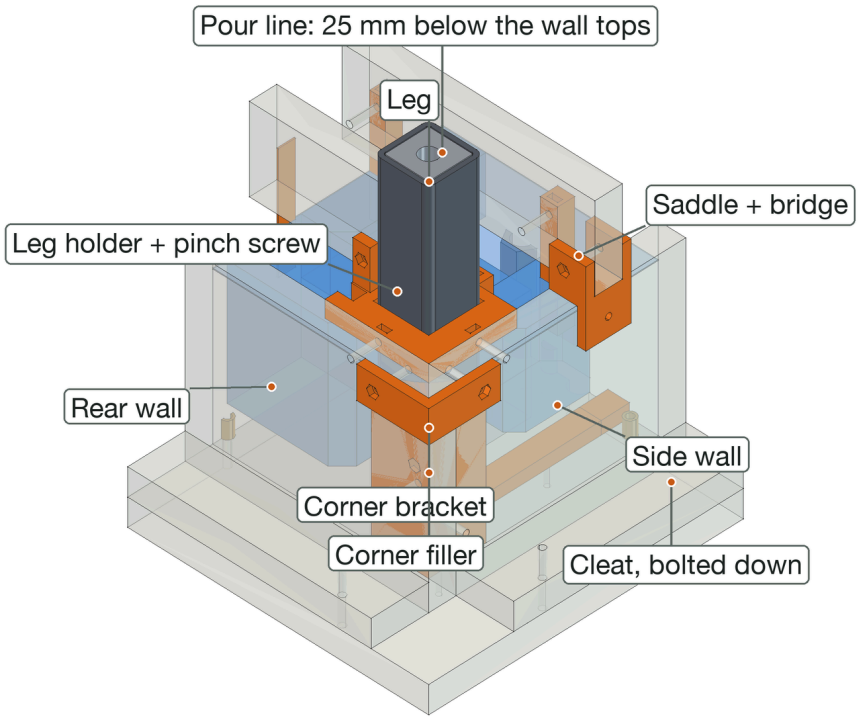
The build guide covers the casting itself (mix, pour, cure); this guide is everything before and after that. All the sizes and hole positions come from the model (`model/build_cnc.py`), so if you change the design, re-run it and the guides, hole maps and printed parts all follow.



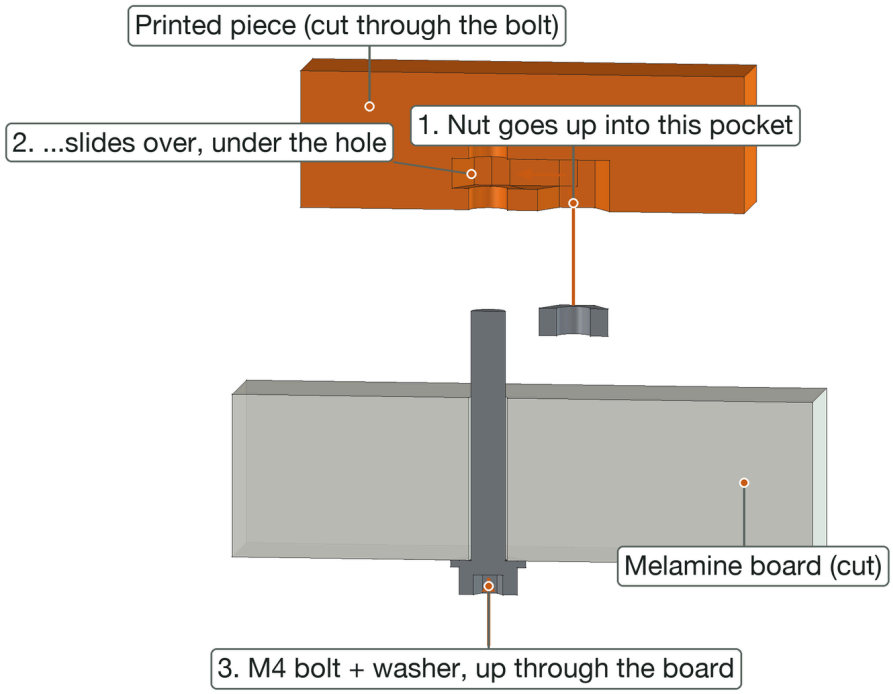
The order of work: print the kit (start first: it is days of printing) → cut and drill the boards → cut the foam → wax → assemble the mold → cast → take it apart. The castings go in this order: base rear half, base front half (against the rear half), the two uprights together, the beam.

1. How the molds work

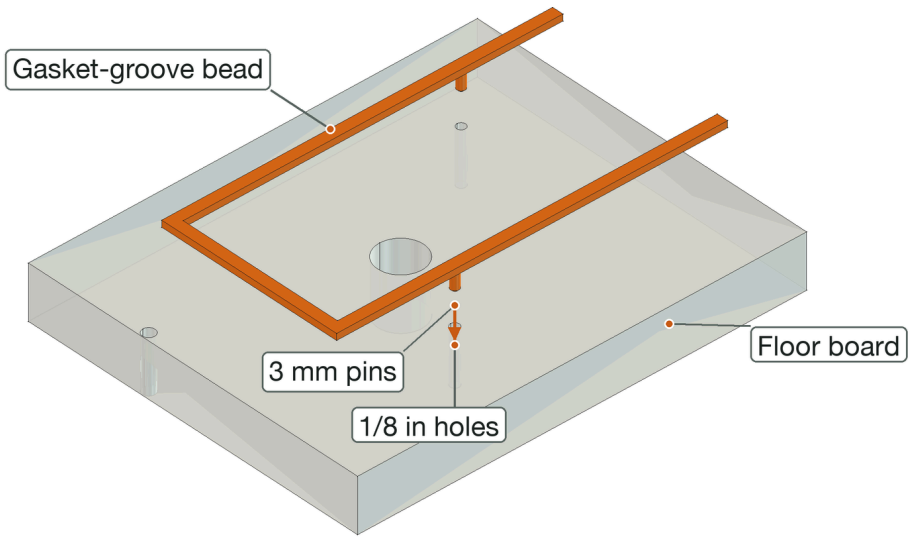
Walls stand 25 mm above the pour line. All the bolts that hold the box together go through that band, so none passes through a face that gets cast against. Fill to the line, not to the top. Strips of board (cleats) bolted down along the outside of each wall's foot keep the walls from moving out; printed corner brackets hold the corners square at the top.



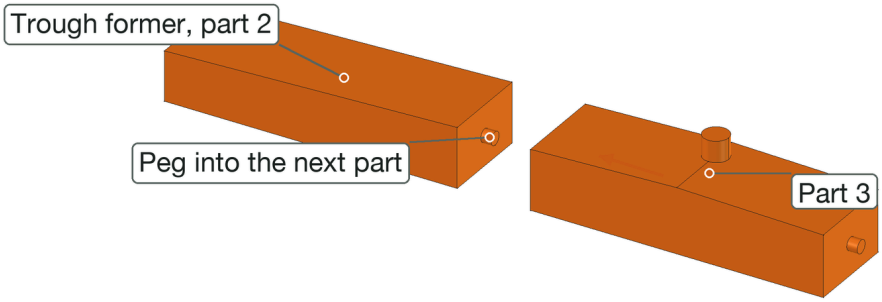
Printed pieces bolt on from outside. Each piece that forms part of the casting's surface (the formers and fillers) has a nut built into it: the nut drops into a pocket in the face that sits against the board, slides sideways under the bolt hole, and an M4 bolt (M3 on thin pieces) goes through the board from outside into it. The pocket is on the board side, so nothing opens onto a cast face, and the bolt comes out from outside after casting. Fit the nuts before the pieces go on.



Thin pieces have pins. The gasket-groove beads and the motor relief are too thin for a nut; their 3 mm pins push into 1/8 in holes. The casting presses them onto the board.



Pieces longer than the printer bed are printed in parts with a peg on one cut face and a socket in the other. Push them together; each part has its own bolts to the board, so they need no glue.



Fits are 0.15 mm a side: snug, but they go together by hand. Board thickness matters: the saddles and cradles are drawn for 19.05 mm board (Home Depot's 3/4 in white melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 19 mm (3/4 in)	about 0.5 m2	all the molds come out of two 4 x 8 ft sheets (layout in molds-complete.pdf)
PETG	0.15 kg	this mold's printed parts (sliced); all the molds and machine parts fit in two 1 kg rolls
Paste wax (Partall #2) and PVA	1 each	release, 3-5 coats
3/8 in quarter-round molding (PVC)	a few meters	the 10 mm outside edges of the castings

Hardware (zinc-plated is fine; grease anything that touches the mix). Buy about 10 % spare of the small bolts, nuts and washers.

Item	Qty
M4 nut	26
M4 washer	26
M4 x 30 bolt	12
M4 x 45 bolt	10
M4 x 70 bolt	4
M5 x 20 screw, greased (holds an insert)	8
M6 x 20 screw, greased (holds a pad)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 198 mm	8

Tools: a drill with sharp brad-point bits (sizes in section 5); a square and a long straightedge; a few clamps; a 7 mm nut driver and a 3 mm hex key; a marker that writes on melamine.

3. Print the parts

14 prints, 0.15 kg of PETG and about 12 hours of printing, sliced in Bambu Studio for an A1 mini (settings in the print guide). Print one corner bracket and one saddle first and check them on an offcut of your board: the saddle should drop over it with a little play.

Folder	File	Qty	g (all)	Time each
beam	`kit-corner-bracket.stl`	4	33	40m 3s
beam	`filler-back-edge-part1of4.stl`	1	19	1h 14m 46s
beam	`filler-back-edge-part2of4.stl`	1	19	1h 14m 46s
beam	`filler-back-edge-part3of4.stl`	1	19	1h 14m 46s

Folder	File	Qty	g (all)	Time each
beam	`filler-back-edge-part4of4.stl`	1	19	1h 15m 1s
beam	`kit-tube-locator.stl`	4	39	55m 37s
beam	`kit-dowel-stop.stl`	2	2	6m 51s

- **Nuts:** press M4 nuts into the hex pockets of the brackets and saddles; slide the nuts into the pockets of the formers and fillers (a dab of grease keeps them from falling out). A nut on a bolt through the hole pulls a tight one into place.
- **Pegs:** test-fit the parts of each long former; a light pass with a file if a peg is tight.

4. Cut the boards

Most boards are strips, so the quickest way is to have the store rip each sheet into strips at the widths below, and crosscut them at home. Label every board with its name on the outside face as you cut it, and mark which face is the cast face (the melamine face that the mix touches).

Board	Qty	Size (mm)	Notes
Beam floor	1	648 x 258	
Beam top wall	1	568 x 165	
Beam bottom wall	1	568 x 165	
Beam end wall, left	1	140 x 165	
Beam end wall, right	1	140 x 165	
Beam cleat, top	1	568 x 40	offcut strip, flat on the floor
Beam cleat, bottom	1	568 x 40	offcut strip, flat on the floor
Beam cleat, left	1	140 x 40	offcut strip, flat on the floor
Beam cleat, right	1	140 x 40	offcut strip, flat on the floor

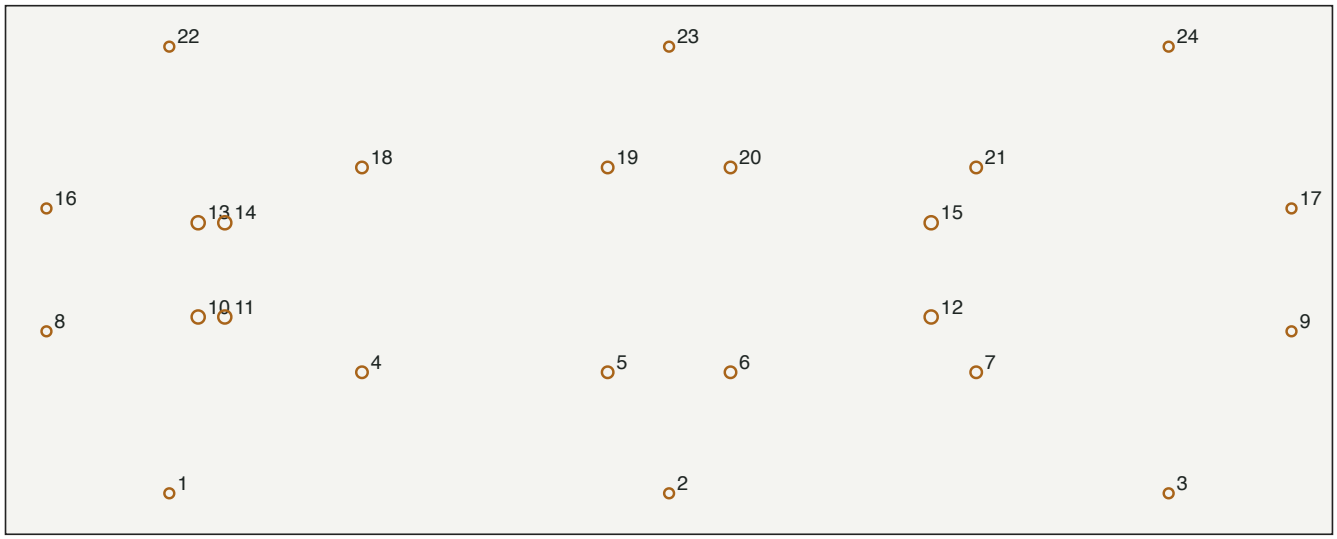
5. Drill the boards

- Drill every hole from the **cast side**, so any tear-out is on the outside. Clamp a scrap under the board where the bit comes out.
- The drawings and tables are seen from the cast side: x from the left edge, y from the bottom edge. For a wall the bottom edge sits on the floor; for a floor, the bottom of the drawing is toward the front wall (the beam's bottom wall). Left and right are as you look at the cast face, so a floor is mirrored compared with the machine. Bridges and cleats are drawn from the top.
- The brackets and saddles are drill guides: clamp one in place and drill through its holes.
- A washer goes under every bolt head; countersink nothing.

Hole	Drill	For
4.5 mm	11/64 in	back-edge, cleat, corner bracket, tube locator
5.5 mm	7/32 in	M5 insert screw

Hole	Drill	For
6.5 mm	17/64 in	M6 insert screw
9.6 mm	25/64 in	3/8 dowel
13.1 mm	33/64 in	1/2 core rod

Beam floor



0,0 · 648 x 258 mm · seen from the cast side

#	x	y	Hole	For
1	80	20	4.5	cleat
2	324	20	4.5	cleat
3	568	20	4.5	cleat
4	174	79	5.5	M5 insert screw
5	294	79	5.5	M5 insert screw
6	354	79	5.5	M5 insert screw
7	474	79	5.5	M5 insert screw
8	20	99	4.5	cleat
9	628	99	4.5	cleat
10	94	106	6.5	M6 insert screw
11	107	106	6.5	M6 insert screw
12	452	106	6.5	M6 insert screw
13	94	152	6.5	M6 insert screw
14	107	152	6.5	M6 insert screw
15	452	152	6.5	M6 insert screw
16	20	159	4.5	cleat
17	628	159	4.5	cleat

#	x	y	Hole	For
18	174	179	5.5	M5 insert screw
19	294	179	5.5	M5 insert screw
20	354	179	5.5	M5 insert screw
21	474	179	5.5	M5 insert screw
22	80	238	4.5	cleat
23	324	238	4.5	cleat
24	568	238	4.5	cleat

Beam top wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	37	115	13.1	1/2 core rod
6	81	115	13.1	1/2 core rod
7	487	115	13.1	1/2 core rod
8	531	115	13.1	1/2 core rod
9	84	133	4.5	back-edge
10	219	133	4.5	back-edge
11	351	133	4.5	back-edge
12	484	133	4.5	back-edge
13	33	154	4.5	corner bracket
14	535	154	4.5	corner bracket

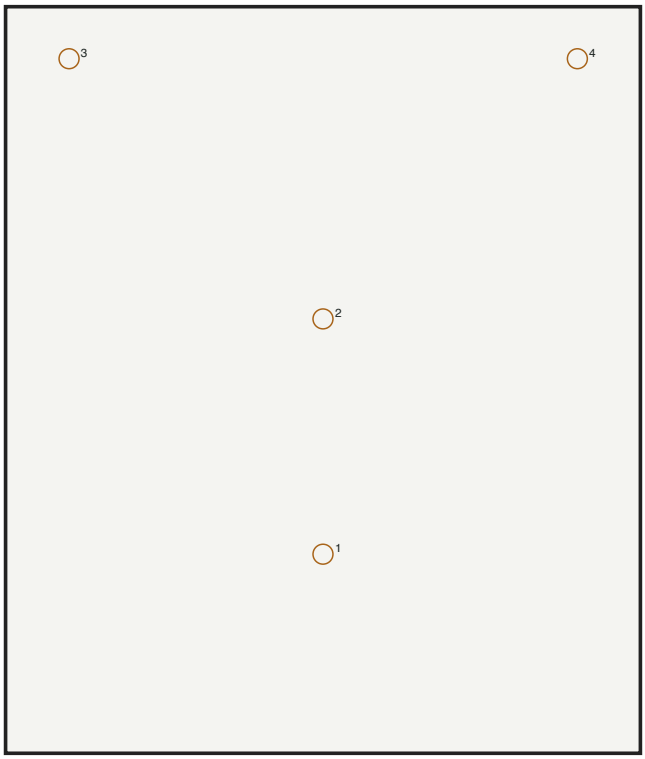
Beam bottom wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	509	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod
9	487	115	13.1	1/2 core rod
10	531	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	535	154	4.5	corner bracket

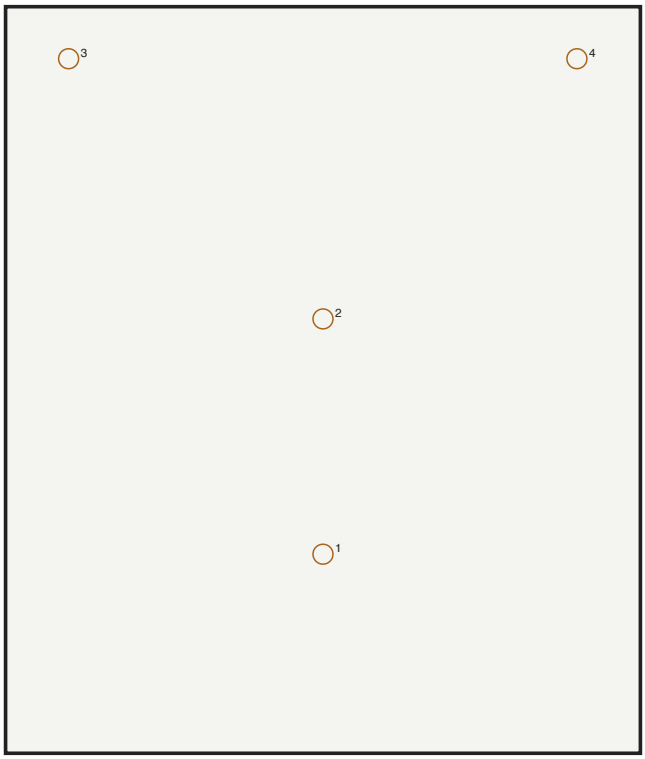
Beam end wall, left



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam end wall, right



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam cleat, top



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, bottom



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, left



0,0 · 140 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat

Beam cleat, right



0,0 · 140 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat

6. Prepare the surfaces

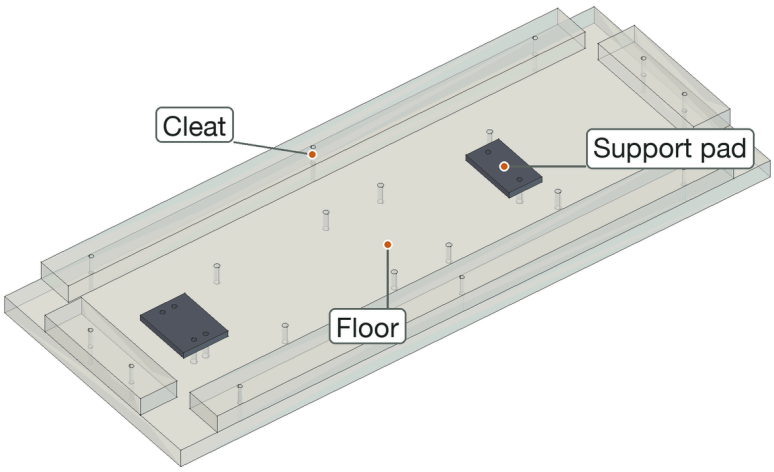
- Wax every face the mix touches, boards and removable printed pieces alike: 3-5 thin coats of paste wax, each buffed, then a coat of PVA. Don't wax the pieces that stay in (core posts, cast-in parts): scuff them and wipe them with acetone so the epoxy grips.
- The castings' outside edges are rounded 10 mm: glue 3/8 in quarter-round molding into the inside corners that form outside edges (floor to wall), and run a small silicone fillet into the others.
- Grease every screw, bolt and rod that will come out of the casting.

7. Put the mold together

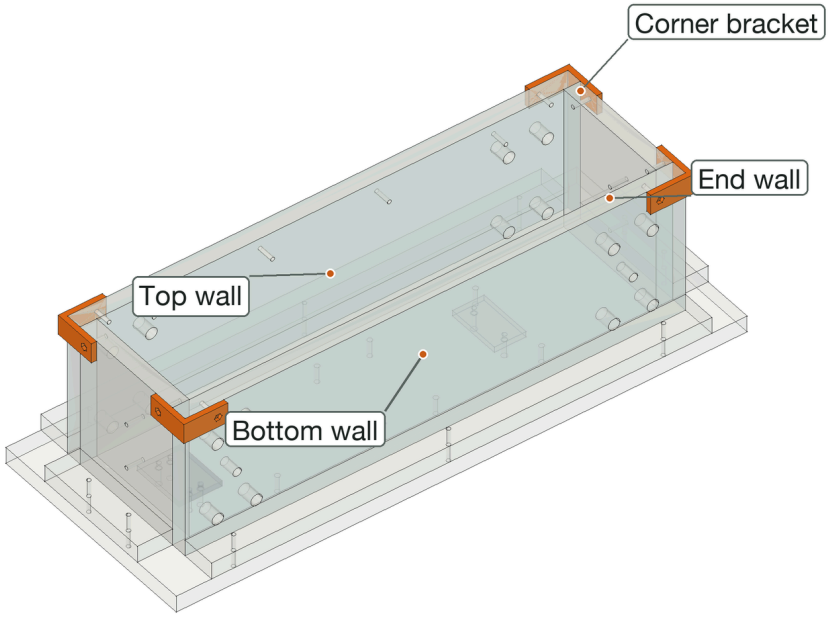
Beam

The beam is cast with its front (rail) face down on the floor and its back open.

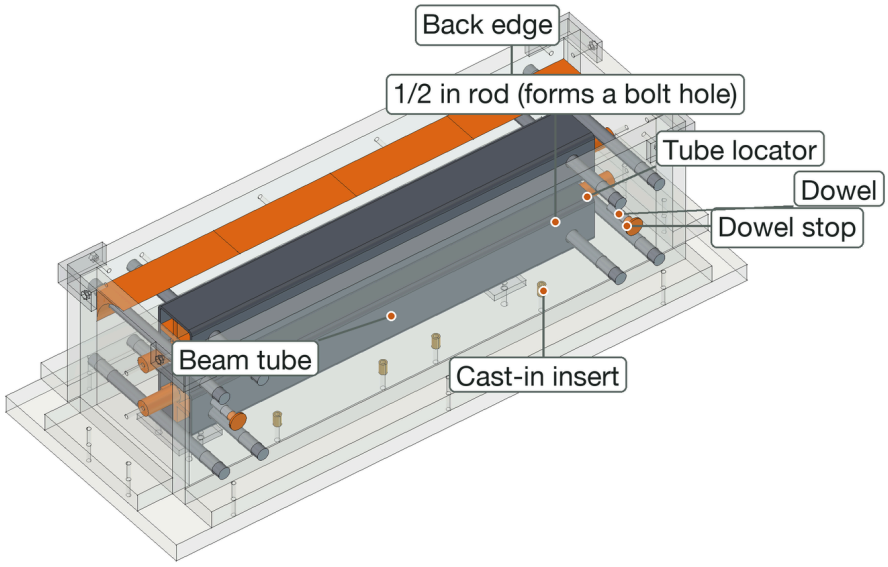
1. Bolt the cleats down; screw the support pads and riser inserts to the floor on greased screws.

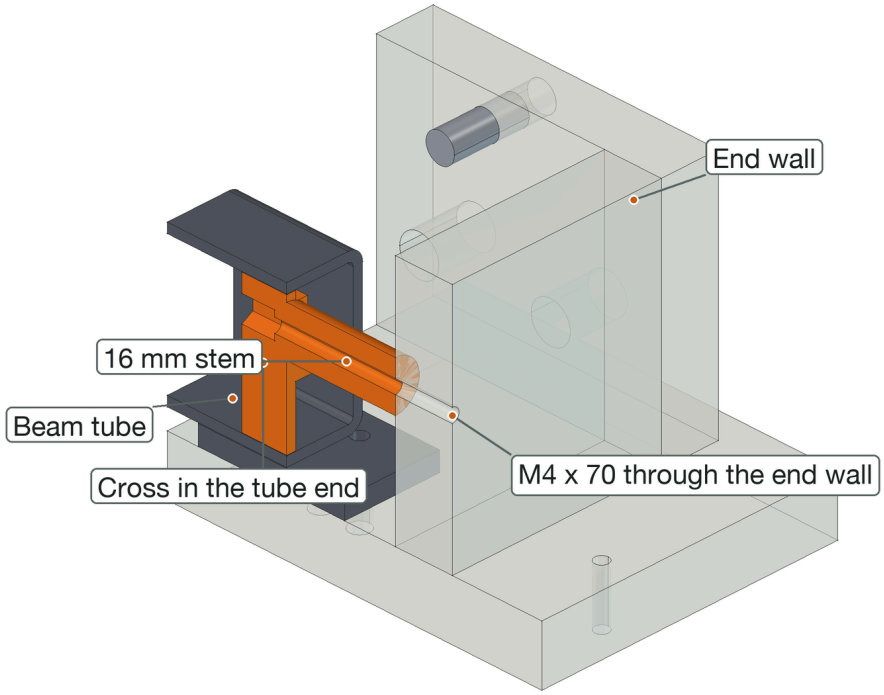


2. Stand the walls against the cleats and fit the four corner brackets.



3. Bolt the tube locators through the end walls (M4 x 70 into the nut in each cross) and hang the tubes between them. Bolt the back-edge filler to the top wall, push the dowels into the bottom wall onto their printed stops, and slide the greased 1/2 in rods through both walls for the eight bolt holes. Fill to the line.





8. Pouring, and the pour line

- The pour line is 25 mm below the wall tops. Mark it inside each wall with a pencil line, and screed the top with a straight board that has two 25 mm feet riding on the wall tops.
- Vibrate each layer until it stops bubbling; the brackets and cleats hold the walls, so you can vibrate the floor board hard. Check the corner brackets and saddle bolts are still tight halfway through.
- Keep the mix off the bridges and saddles; scrape it off before it cures.

9. Taking it apart, and reuse

1. Leave it the full cure time (build guide). Unbolt the bridges and saddles, then the corner brackets, then the cleats; pull the walls straight off (a plastic wedge in the corner helps).
2. Unscrew every insert screw, coupler bolt and rod from outside, then lift the casting off the floor (or the floor off the casting).
3. Unbolt and pull the formers and fillers (the bolts come out from the board side first). A wooden dowel and a light tap frees a stuck one.
4. Scrape the boards, re-wax them, and store the kit pieces with their nuts in a box per mold. The same kit does the next casting; only the formers and pieces that stay in are per casting.

10. If something doesn't fit

Problem	Fix
A nut won't slide under the hole	Clean the pocket with a nut on a bolt; a warm nut (hair dryer) pushes in

Problem	Fix
A bracket or saddle is loose on the board	Your board is thinner than set: shim with tape, or set `mold_board` and reprint
A pin won't go into its hole	Run a 1/8 in bit through the hole again; trim the pin tip with a knife
A foam core moves while pouring	Check its post bolt is tight and the post is glued; add carpet tape under it
Mix leaks at a corner	A silicone fillet inside the corner before waxing; check the diagonals and brackets
A former is stuck after curing	Bolts out first; then a dowel and a light tap from its thick end